



24ADC001- BUSINESS INTELLIGENCE

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**2026-2027
LAB RECORD**

**DEPARTMENT OF ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE**

INDEX

Expt. No	Title of the Experiment	Page No	Total Marks	Faculty Signature
1.	Explore the interface and basic features of the BI tool			
2.	Load and visualize a sample dataset.			
3.	Import a dataset into the BI tool. And Cleanse data by handling missing values, outliers, and inconsistencies.			
4.	Transform data to suit BI reporting requirements and Design a dashboard with key performance indicators (KPIs).			
5.	Develop interactive dashboards for dynamic data exploration.			
6.	Integrate data from various sources for comprehensive analysis			
7.	Implement advanced chart types (treemaps, heatmaps, etc.).			
8.	Apply BI tools for forecasting and predictive analytics.			

Average marks

Faculty Signature

EXPT NO:1	EXPLORE THE INTERFACE AND BASIC FEATURES OF BI Tool
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the purpose of Business Intelligence (BI) tools like Tableau and Power BI?

Business Intelligence tools such as Tableau and Power BI help organizations turn raw data into meaningful information. They make it easier to identify patterns, trends, and business performance through visual reports. Instead of analyzing large spreadsheets manually, users can understand data quickly using charts and dashboards. These tools also support better planning and faster decision-making. They are widely used in finance, marketing, healthcare, and many other industries. Overall, BI tools help businesses make informed decisions based on reliable data.

2. List any three common data sources that Tableau and Power BI can connect to.

Tableau and Power BI support connections to many different types of data sources. Some of the most commonly used ones are Microsoft Excel files, CSV (Comma-Separated Values) files, and SQL databases such as MySQL or SQL Server. They can also connect to cloud services like Google Sheets and Azure. This flexibility allows businesses to combine data from multiple sources into one report. As a result, users can perform better analysis without manually merging datasets.

3. What is a dashboard in BI, and how is it different from a single chart?

A dashboard is a collection of multiple visualizations displayed together on one screen. It provides a complete overview of important business information in a single place. A single chart focuses on only one aspect of the data, such as sales or profit. In contrast, a dashboard combines charts, tables, KPIs, and filters to present a broader picture. Users can interact with dashboards to explore data in more detail. This makes dashboards more useful for business monitoring and decision-making.

4. Define the following terms:

a) Visualization

Visualization is the process of representing data using graphical elements such as charts, graphs, maps, and tables. It helps users understand complex information more easily than reading raw numbers. Good visualizations highlight important trends, comparisons, and patterns in the data. They also make reports more attractive and easier to interpret. Business Intelligence tools rely heavily on visualization for effective communication. This enables faster and more accurate business decisions.

b) KPI

A KPI, or Key Performance Indicator, is a measurable value used to evaluate how well a business or process is achieving its goals. It helps organizations monitor

their performance over a specific period. Examples include total sales, profit margin, customer satisfaction, or revenue growth. KPIs allow managers to identify strengths and areas that need improvement. They also support strategic planning and performance tracking. In BI dashboards, KPIs are often displayed as highlighted summary values.

c) Data Filter

A data filter is a feature that allows users to display only the information they need from a dataset. Instead of viewing all available records, filters help narrow the data based on specific conditions such as date, product, or location. This makes analysis more focused and meaningful. Filters also improve report readability by removing unnecessary information. In Tableau and Power BI, filters are commonly used to explore data from different perspectives. They make dashboards more interactive and user-friendly..

5. Why is it important to clean and transform data before visualization?

Cleaning and transforming data ensures that the information used for analysis is accurate and reliable. Raw data often contains missing values, duplicate records, and formatting errors that can lead to incorrect results. By correcting these issues, the visualizations become more meaningful and trustworthy. Data transformation also helps organize the data into a suitable format for analysis. This improves the quality of reports and reduces the chances of making poor business decisions. Clean data is the foundation of effective Business Intelligence.

6. What kind of chart would you use to analyze stock price trends over time?

A line chart is the most suitable visualization for analyzing stock price trends over time. It clearly shows how stock prices increase or decrease across different dates. The continuous line makes it easy to identify patterns, fluctuations, and long-term movements. Investors and analysts often use line charts to compare historical performance. Multiple stock prices can also be displayed on the same chart for comparison. This makes trend analysis simple and effective.

7. Explain the role of slicers in Power BI and filters in Tableau.

Slicers in Power BI and filters in Tableau help users interact with reports by selecting specific data to display. They allow users to focus on information based on categories such as dates, products, or regions. This makes reports more flexible and easier to explore. Instead of creating multiple reports, one dashboard can display different views using these features. They also improve the user experience by making data exploration simple. Overall, they support faster and more detailed analysis.

8. What does a calculated field or measure do in BI tools?

A calculated field or measure allows users to create new values by applying formulas to existing data. It helps perform calculations such as profit, percentage growth, average sales, or total revenue without changing the original dataset. These calculations are updated automatically whenever the data changes. This saves time and improves reporting accuracy. Calculated fields also make dashboards more informative by providing additional business insights. They are an important feature in both Tableau and Power BI.

9. What are the advantages of using interactive dashboards for financial data analysis?

Interactive dashboards help users analyze financial data quickly and efficiently. They allow users to filter, compare, and explore information without creating separate reports. Financial trends, profits, expenses, and stock performance can be monitored in real time. This improves decision-making by providing clear and updated insights. Interactive dashboards also reduce manual analysis and make complex data easier to understand. As a result, businesses can respond faster to financial changes and opportunities.

10. Which function would you use to find the absolute difference between predicted and actual stock prices?

The **ABS()** function is used to calculate the absolute difference between predicted and actual stock prices. It returns the positive value of the difference, regardless of whether the prediction is higher or lower than the actual price. For example, **ABS(Predicted Price - Actual Price)** gives the prediction error without showing a negative value. This makes it easier to measure forecasting accuracy. The **ABS()** function is commonly used in Power BI, Tableau, and Excel for error analysis. It helps analysts evaluate the performance of prediction models.

IN-LAB EXERCISE:

Objective:

To understand the basic interface, data import process, and fundamental visualization capabilities of **Tableau** and **Power BI** using a stock prediction dataset, and to explore their usefulness in analyzing financial data.

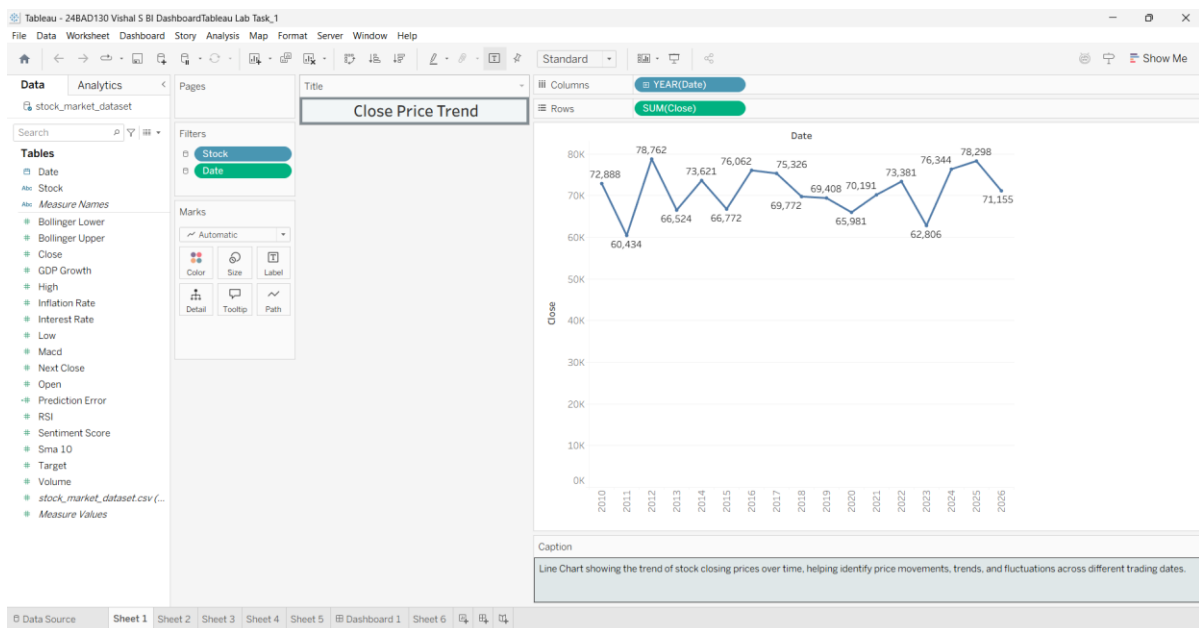
Tools Required:

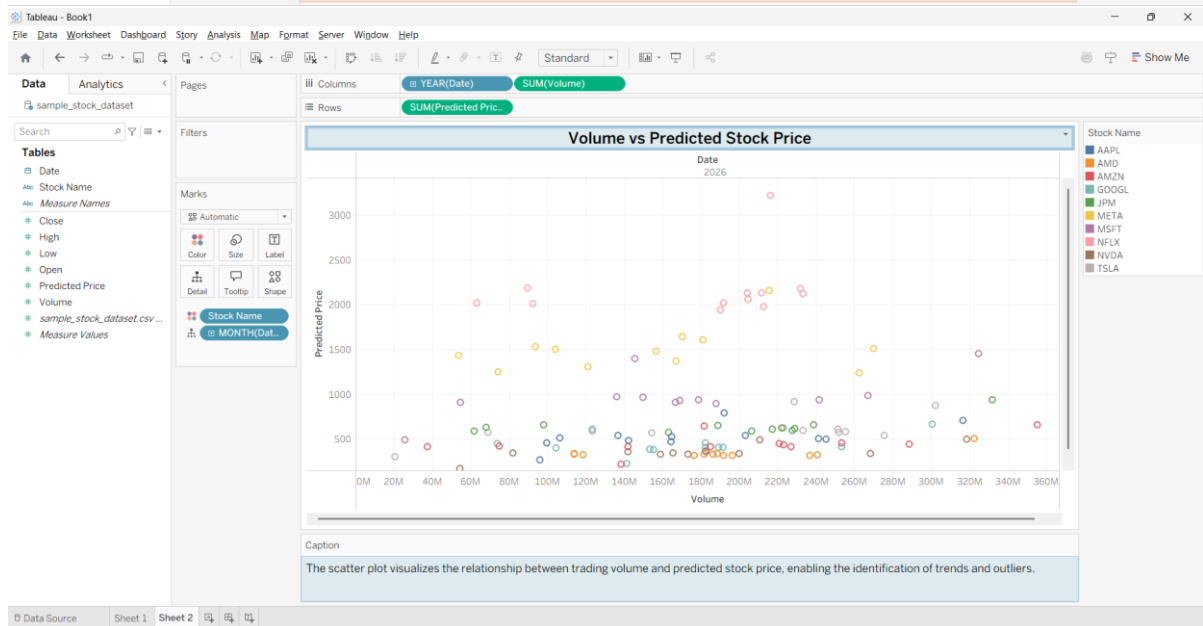
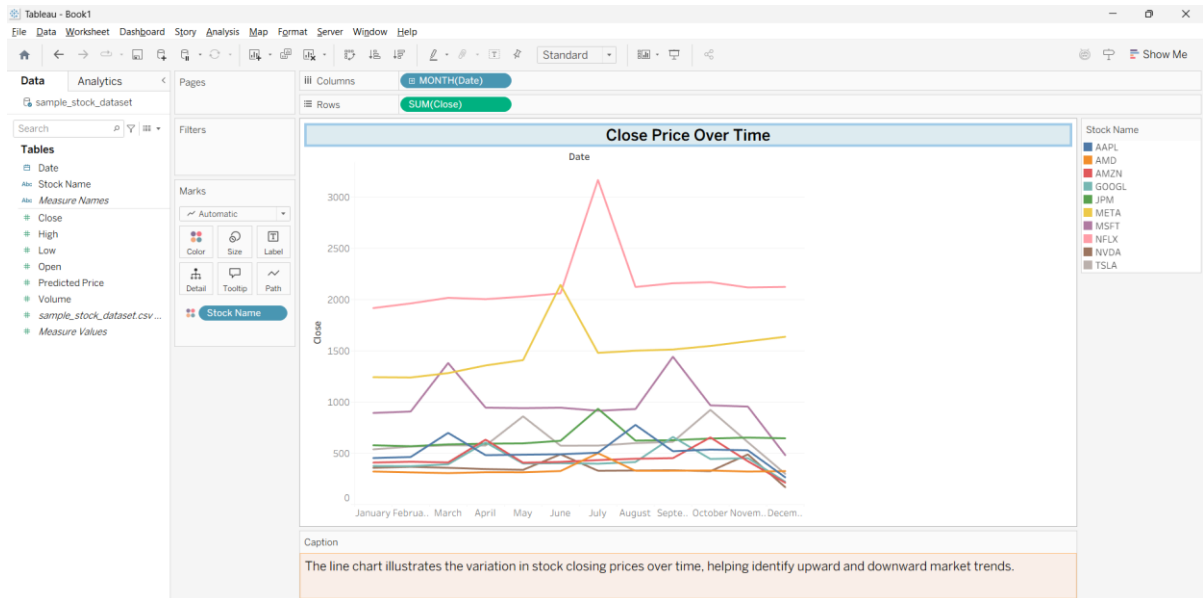
- Tableau Public / Tableau Desktop
- Power BI Desktop
- Sample Dataset (CSV/Excel) containing:
Date, Stock Name, Open, Close, High, Low, Volume, Predicted Price

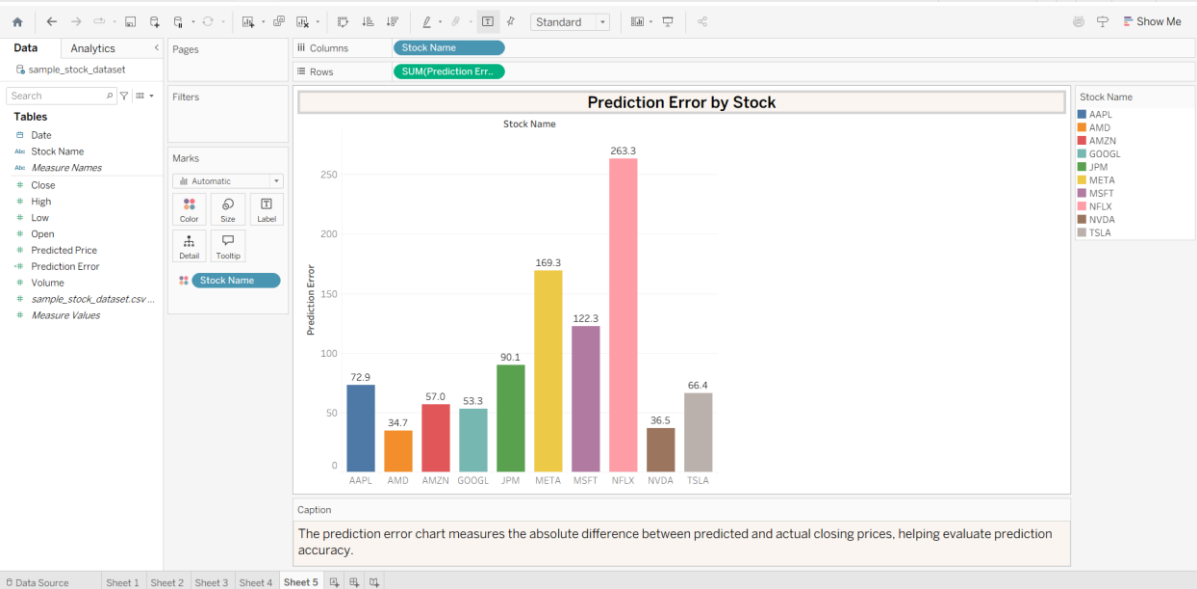
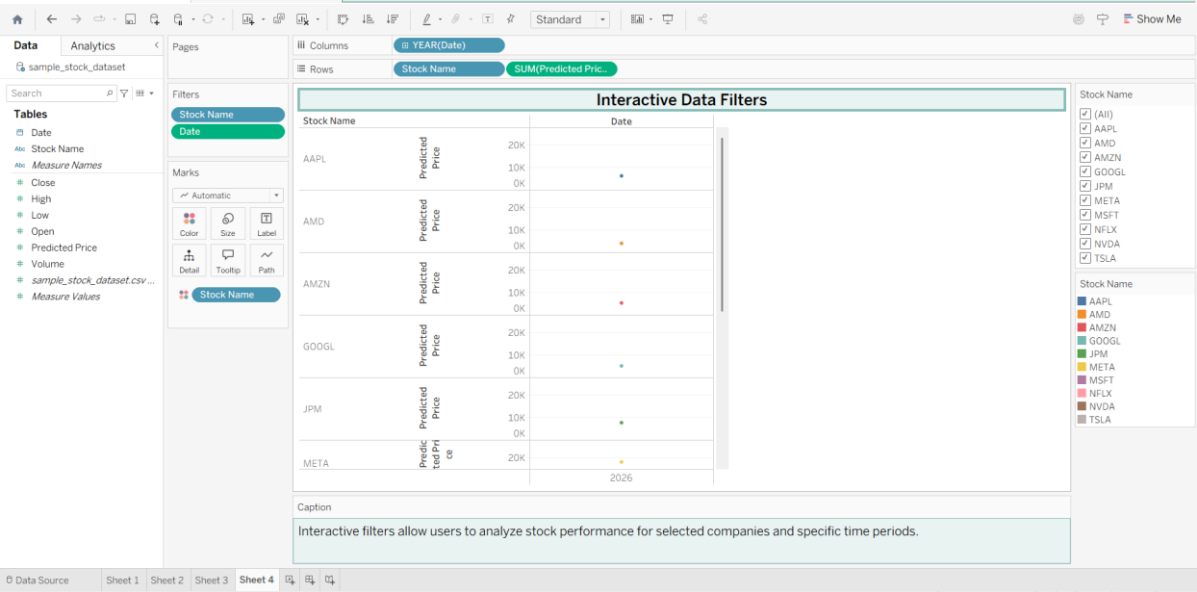
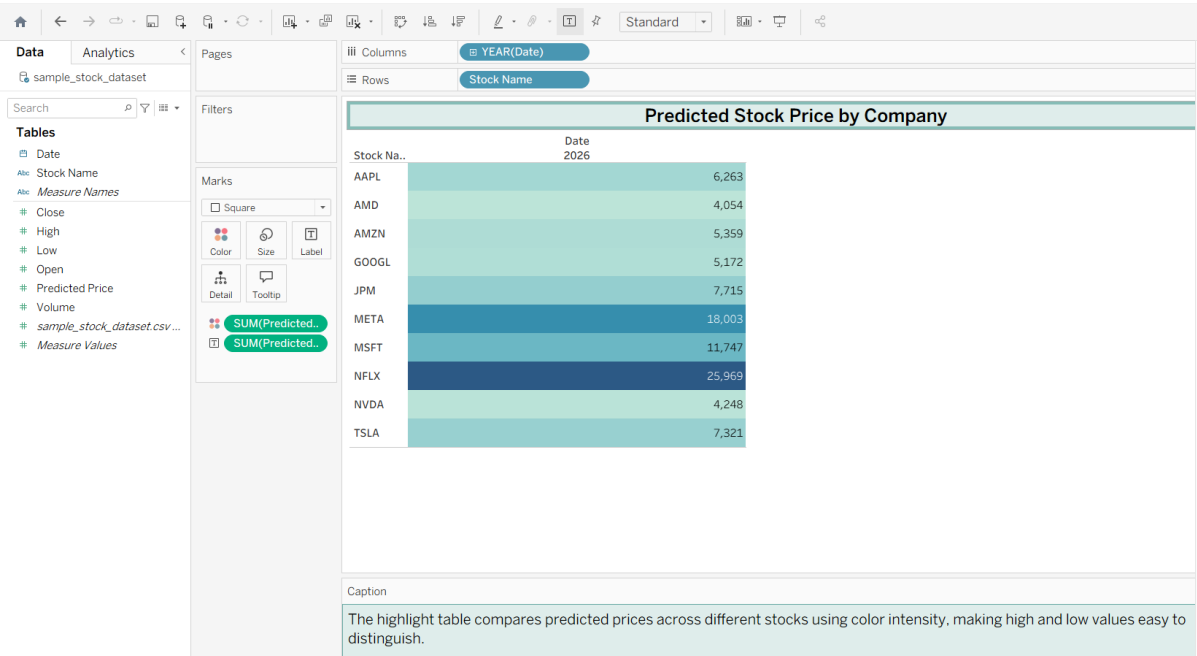
Exercise Steps:

Part 1: Working with Tableau

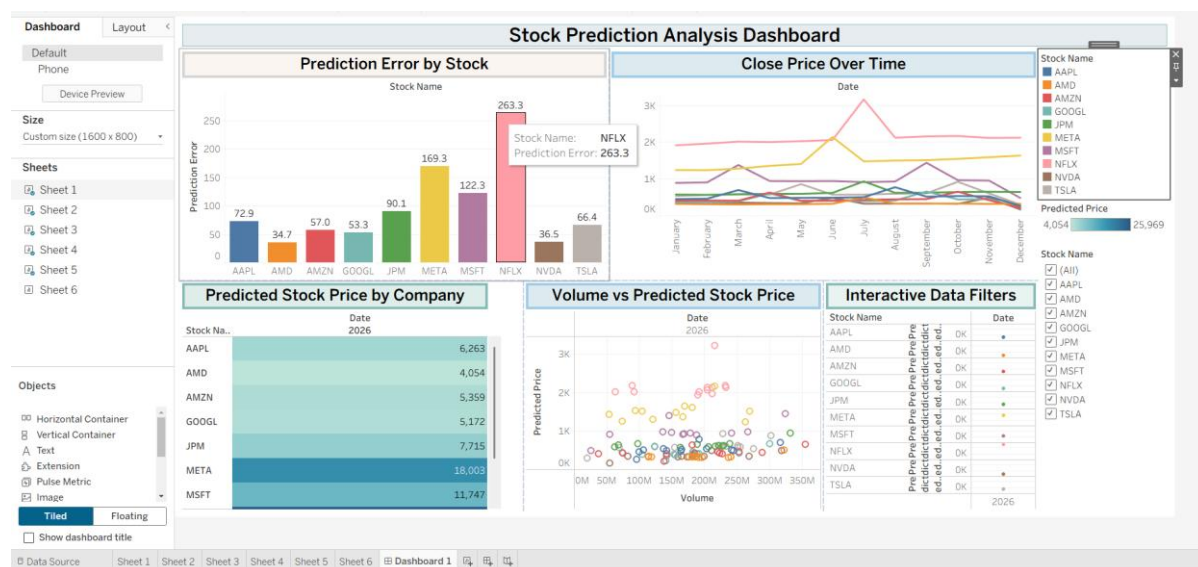
1. **Launch Tableau** and connect to the stock prediction dataset.
2. **Explore the Interface:** ○ Understand components like **Data Pane**, **Sheets**, **Marks Card**, and **Dashboards**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Show Close Price over time. ○ **Scatter Plot:** Plot Volume vs. Predicted Price.
 - **Highlight Table:** Display Predicted Price by stock.
4. **Apply Filters:**
 - Add filters for Stock Name and Date Range.
5. **Create Calculated Field:** ○ Calculate prediction error as $|\text{Predicted} - \text{Close}|$.
6. **Design a Basic Dashboard:**
 - Combine your charts into one interactive dashboard.





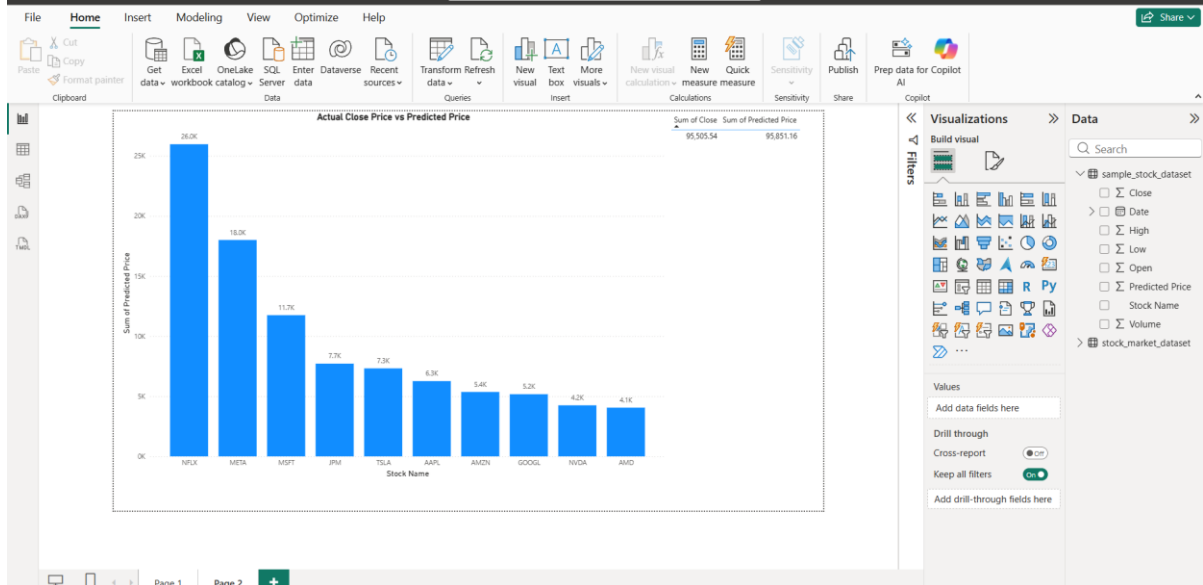
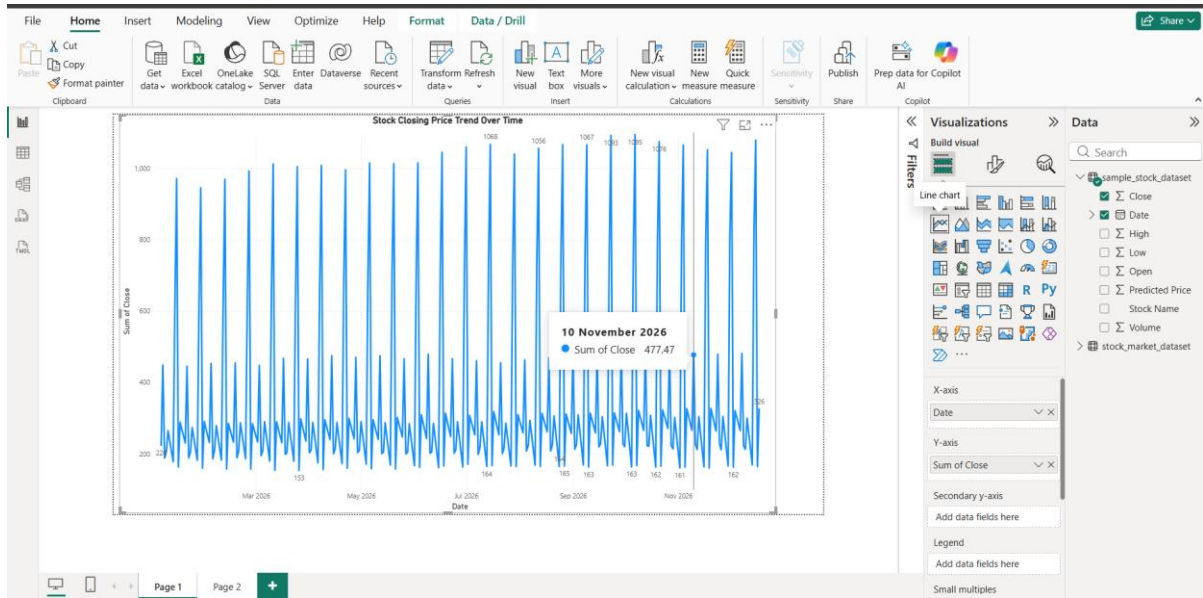


Final dashboard using Tableau:



Part 2: Working with Power BI

1. **Open Power BI Desktop** and import the same dataset.
2. **Explore the Interface:**
 - o Familiarize with **Fields Pane**, **Visualizations Pane**, **Report View**, **Data View**.
3. **Create Basic Visualizations:**
 - o **Line Chart:** Trend of Close Price over time.
 - o **Bar Chart:** Compare Predicted Price vs. Close Price.
 - o **Card (KPI):** Display average prediction error.
4. **Use Filters and Slicers:**
 - o Filter data based on Stock Name and Date.
5. **Create a Simple Measure** using DAX:
 - o Example: $\text{Prediction Error} = \text{ABS}(\text{Predicted} - \text{Close})$
6. **Build a Report Page:**
 - o Arrange visuals in a single interactive page.



Average Prediction Error

3.86

Prediction Error



Interactive Filters and Slicers

Stock Name

- ☐ AMZN
- ☐ GOOGL
- ☐ JPM
- ☐ META
- ☐ MSFT
- ☐ NFLX
- ☐ NVDA
- ☐ TSLA

Sum of Volume by Stock Name



Average Prediction Error

4.89

Prediction Error



Interactive Filters and Slicers

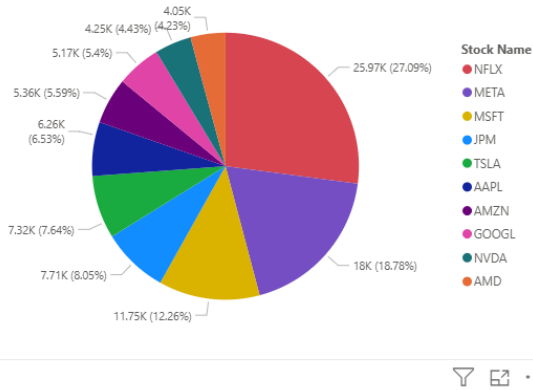
Stock Name

- ☐ AMZN
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- ☒ MSFT
- ☐ NFLX
- ☐ NVDA
- ☐ TSLA

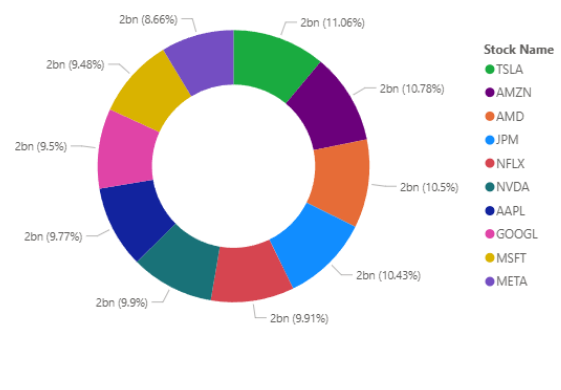
Sum of Volume by Stock Name



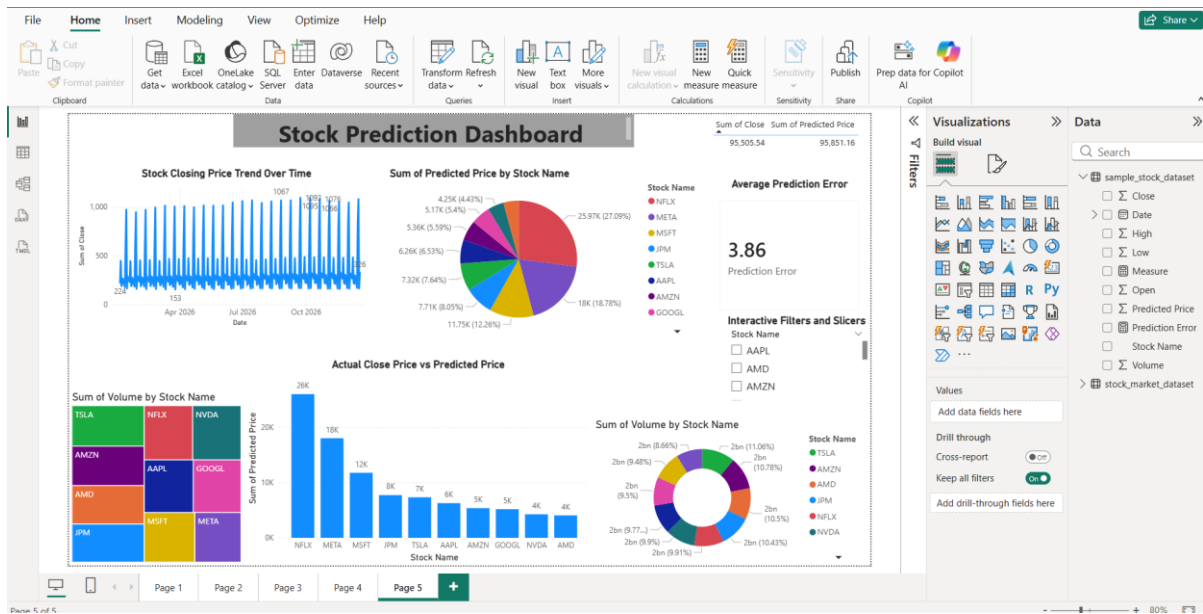
Sum of Predicted Price by Stock Name



Sum of Volume by Stock Name



Final Dashboard Using PowerBI:



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Describe your experience with the Tableau interface. What components did you interact with?

My experience with Tableau was positive because the interface was simple and easy to understand after exploring it for a while. I used the **Data Pane** to access the dataset fields and the **Rows** and **Columns** shelves to create different visualizations. The **Marks Card** helped me customize colors, labels, and chart styles. I also worked with the **Filters** section to analyze specific stocks and dates. Finally, I used the **Dashboard** feature to combine multiple charts into a single interactive view. This made the overall analysis more organized and user-friendly.

2. How did you import the dataset into Tableau and Power BI?

I imported the stock market dataset into Tableau by selecting the Text File option and loading the CSV file. In Power BI, I clicked Get Data, selected the same CSV file, and loaded it into the report. Both tools automatically displayed the available columns and detected the data types. I verified the imported data before creating any visualizations. Since my dataset did not contain a Predicted Price column, I used the Next_Close column for the required analysis. The import process was smooth in both tools.

3. What types of visualizations did you create in both tools? List at least two per tool.

In Tableau, I created a **line chart** to show the closing price trend, a **scatter plot** to compare trading volume with the next closing price, a **highlight table**, and a **bar chart** for prediction error. In Power BI, I created a **line chart**, a **clustered bar chart**, a **donut chart**, a **scatter chart**, and a **KPI card**. Each visualization highlighted a different aspect of the stock data. Together, they made it easier to compare stock performance and understand market trends.

4. How did you calculate the prediction error, and how was it displayed in your dashboard?

Since the dataset did not include a **Predicted Price** column, I treated **Next_Close** as the predicted value. I calculated the prediction error using the formula $ABS(Next_Close - Close)$. In Tableau, I created a calculated field, while in Power BI I created a DAX measure for the same calculation. The prediction error was displayed as a **bar chart**, a **KPI card** showing the average error, and a **table** for detailed comparison. This helped me quickly evaluate the difference between the current and next closing prices.

5. Which filtering or slicing techniques did you apply, and what insights did they reveal?

I applied **Stock Name** and **Date** filters in Tableau and used **Stock** and **Date slicers** in Power BI. These controls allowed me to focus on individual companies and specific time periods without changing the original dataset. By using the filters, I could compare the performance of different stocks and observe changes in closing prices over

time. They also made the dashboard interactive and easier to explore. This helped me identify trends and compare stock behavior more effectively.

6. Compare the process of creating a line chart in Tableau vs. Power BI. What differences did you notice?

In Tableau, I created the line chart by dragging the **Date** field to the Columns shelf and the **Close** field to the Rows shelf. In Power BI, I first selected the **Line Chart** visual and then assigned the Date and Close fields to the appropriate sections. Tableau offered more flexibility in designing the chart through drag-and-drop features. Power BI followed a more structured approach using the Visualizations pane. Although both tools produced similar charts, I found Power BI slightly easier for beginners.

7. How did you organize the dashboard in Tableau and Power BI? What key elements did you include?

I arranged the dashboards so that the most important information appeared at the top for quick understanding. The line chart showing the stock price trend was placed prominently, followed by the bar chart, scatter chart, and highlight table or donut chart. I also included a KPI card to display the average prediction error. Filters and slicers for **Stock** and **Date** were placed on the side for easy interaction. This layout kept the dashboard clean, organized, and simple to navigate.

8. Which tool (Tableau or Power BI) did you find more user-friendly for financial data visualization, and why?

Among the two tools, I found **Power BI** slightly more user-friendly for this experiment. Its interface was familiar, and creating charts, KPI cards, and slicers required fewer steps. Organizing the dashboard and formatting the visuals was also straightforward. Tableau provided more flexibility for creating interactive visualizations, but it took a little more time to learn. Both tools were effective for financial analysis, but Power BI felt easier for beginners.

9. What challenges did you face while creating visualizations or dashboards?

One of the main challenges was that my dataset did not contain the **Predicted Price** column mentioned in the lab manual. To complete the exercise, I replaced it with the **Next_Close** column and modified the prediction error formula. I also spent time selecting suitable visualizations and arranging them neatly on the dashboard. Ensuring that all filters and slicers interacted correctly with every chart required some adjustments. These challenges improved my practical understanding of Business Intelligence tools.

10. How can the knowledge gained from this lab be applied to real-world financial data analysis?

The knowledge gained from this lab can be applied to analyze stock market performance and other financial datasets in real business environments. BI tools help convert raw financial data into meaningful dashboards that support faster decision-making. Interactive reports make it easier to monitor stock prices, compare company

performance, and identify market trends. Analysts can use these dashboards to evaluate investments and prepare financial reports. Overall, this lab strengthened my understanding of how Business Intelligence tools are used for practical financial analysis and business decision-making.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		